

Pragya Prakash

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Education

University of Toronto

Sep 2022 – May 2027 (Expected)

Bachelor of Industrial Engineering – Minor in Artificial Intelligence & Business

Toronto, ON

- Relevant CourseWork: Data Science (Python); Artificial Intelligence; Machine Learning; Data Modelling (SQL); Data Structures and Algorithms; OOP(Java); Supply Chain Management; Design Analysis and Information Systems (JS)

Work Experience

Litens Automotive Patnership

May 2025 - Apr 2026

Project Management Intern/ Business Development Team

Vaughan, ON

- Led cross-functional collaboration to restructure the company's JIRA project setup, resulting in a unified Master Board template that improved task visibility and workflow consistency across multiple teams.
- Automated bulk creation of epics and tasks using Python scripts, significantly reducing manual setup time and accelerating project onboarding for the Minihub and M12 Smart Pump initiatives.
- Supported resource allocation and budget tracking using Dynamics 365 (LPM), contributing to improved time management and cost transparency across engineering teams.
- Initiated Power BI dashboard integration to visualize budget and cost metrics, enhancing reporting capabilities for Litens industry entries.
- Assisted in ASPICE process alignment by coordinating with software and firmware teams, helping ensure compliance with automotive software standard

OBOTZ Robotics

April 2024 - October 2024

Robotics Instructor - Embedded C, Scratch

Newmarket, ON

- Taught robotics to students aged 5–14, focusing on Embedded C, sensor integration, and hands-on projects like collision-avoiding robots and automated streetlights.
- Designed hands-on demonstrations (streetlights, loop-wire game, motor-spinning candle, car headlights) to teach resistors, LDRs, and electricity generation, translating abstract theory into applied, real-world understanding.

WeConnect Community Services

July 2018 - August 2021

Community Service Event Organizer

Toronto, ON

- Collaborated with a team to create engaging social media campaigns and promotional content, encouraging seniors and newcomers to join community events and programs.
- Initiated New Year, Easter, and Canada Day events, teaching Excel and Zoom skills during COVID to enhance digital literacy, strengthening leadership and problem-solving abilities, and improving community engagement.

Projects

Mammographic BI-RADS Analysis for Early Breast Cancer Diagnosis | Python

- Preprocessed mammographic imaging and structured health data as part of a 4-person team, enabling development of specialized neural network models for cancer prediction and BI-RADS classification.
- Engineered synthetic data variations (e.g., age shifts, breast density changes, image rotations and color adjustments) to simulate real-world clinical variability, improving model robustness and generalization.
- Achieved 72% accuracy on unseen, modified datasets using a multimodal neural network approach, validating the model's adaptability to diverse patient demographics and imaging conditions.

Emotions Classification using BERT and GPU | Python

- Fine-tuned BERT's transformer architecture for emotion recognition in Twitter text data, achieving a 94% improvement in classification accuracy through contextual language modeling.
- Implemented GPU-based training pipelines with TensorFlow and PyTorch, significantly reducing model training time and optimizing resource utilization for high-performance results.

SkinMatch – Skincare Recommendation Web Application | Java, SQL

- Collaborated in a team of 8 for building a full-stack skincare recommendation platform using Java, SQL, REST APIs, and Appsmith, leading user testing and model validation based on skin data. Delivered a responsive web app with personalized suggestions and robust CRUD features, enhancing accuracy by 94% and user engagement.
- Led user testing by designing and executing test cases to validate recommendation algorithms and user flows, enhancing model precision and usability through iterative testing and structured feedback.

Hospital Readmission Predictive Model | *Python*

- Performed data preprocessing, exploratory analysis, and feature engineering as part of a 4-person team, preparing a dataset of patient records from 30 US hospitals for predictive modeling.
- Developed an XGBoost model achieving 66.9% AUC to predict hospital readmissions for diabetic patients, enabling earlier identification of high-risk cases.
- Simulated hospital operations using the model's predictions to optimize bed utilization and improve discharge planning in Acute and Emergency Wards.

Optimized University Course System Development | *Java, SQL*

- Extended a University Course Management System by redesigning the database schema and integrating new features (Teaching Assistants, course prerequisites) using Java and Hibernate.
- Wrote SQL queries to report on course statistics (grade distributions, enrollments), improving system performance and data accuracy through optimized database design and issue resolution.

Traveling Salesman Problem (TSP) Solver | *Java*

- Applied dynamic programming to build an optimized, accurate data-loading model using Nearest Neighbour (NN), Node Insertion (NI), and Nearest Neighbour First-Last (NFL) algorithms, with extensive debugging for reliability.
- Leveraged object-oriented programming in Java to optimize models, balancing cost and time to improve efficiency and reduce program runtime by 22.5%.

Communication Software App for Peel Patients | *Python*

- Collaborated with a team of 6 to develop Python-based tablet and smartphone applications, enhancing communication between ADHD patients and caregivers through user-friendly interfaces.
- Integrated image-recognition technology to convert images into words and expanded the image database, improving interaction accuracy between patients and caregivers.
- Ensured professional, consistent formatting of technical documents and reports to align with client standards.

Skills

Languages: Python, Java, SQL, JavaScript, HTML, CSS, R, MATLAB, Simio, AutoCAD

Tools: Google Colab, JIRA, MS Projects, Eclipse, VS Code, IntelliJ IDEA, Swish, SPARQL, Microsoft Access, PowerBI

Libraries/Frameworks: Numpy, Pandas, Matplotlib, SpringFramework, Seaborn, Torch, ASPICE

Professional Skills: Teamwork, Problem Solving, Project Management